# *Framingham Longitudinal Stroke Risk Profile*

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```
## ‘summarise()‘ has regrouped the output.

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## i Summaries were computed grouped by SEX and PERIOD.

## i Output is grouped by SEX.

## i Use ‘summarise(.groups = "drop_last")‘ to silence this message.

## i Use ‘summarise(.by = c(SEX, PERIOD))‘ for per-operation grouping

##   (‘?dplyr::dplyr_by‘) instead.
```

## Introduction

This project investigates stroke risk in the Framingham Heart Study, a long-term prospective cohort that has followed adults in Framingham, Massachusetts for cardiovascular outcomes since 1948. For this analysis, we use a subset of 4,434 participants with baseline measurements of cardiovascular risk factors and stroke outcomes. Although participants were followed for up to 24 years, this project focuses on the first 10 years of follow-up to ensure that baseline risk factors remain relevant to the outcome window.

The primary goal is to identify which baseline risk factors predict time to first stroke within 10 years, analyzed separately for males and females. A secondary aim is to estimate 10 year stroke probabilities for several clinically relevant risk profiles. Because risk factors were mea-

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sured at three exam periods, an additional aim is to describe how systolic blood pressure and diabetes status change over time and to discuss how such changes might influence analyses that rely only on baseline values.

## Methods

All analyses will be conducted in R v4.5.2 (2025-10-31 ucrt). Participants with evidence of stroke at baseline will be excluded. Follow-up time will be defined using TIMESTRK and restricted to 10 years. Individuals who experience a non-stroke event before 10 years will be censored at the time of event. Events occurring after 10 years will be censored at 10 years. A new event indicator will be created to capture first stroke within the 10 year window.

Baseline covariates include age, systolic blood pressure, diabetes, smoking status, BMI, cholesterol, coronary heart disease, and blood pressure medication use. Variables will be checked for implausible values and recoded when necessary. Missingness will be summarized descriptively. Complete case analysis will be used for the final Cox models. Analyses will be stratified by sex. Age, systolic blood pressure, and diabetes will be included *a priori*, additional baseline covariates will be evaluated using backwards selection based on AIC. Differences in final models by sex will be noted and interpreted as needed.

Kaplan–Meier curves and log-rank tests will be used to compare stroke-free survival across baseline risk factors. The primary analysis will use Cox proportional hazards models. Proportional hazards assumptions will be assessed using Schoenfeld residuals and log-minus-log survival plots. For each sex, 10 year predicted stroke probabilities will be estimated from the final Cox model for five risk profiles at ages 40, 50, and 60. The risk profiles are: 1. No baseline risk factors, 2. High systolic blood pressure ($\geq$ 160), 3. Diabetes, 4. Both high systolic blood pressure and diabetes, 5. One additional profile chosen based on model results. Although the primary analysis will only use baseline covariates, systolic blood pressure and diabetes will be summarized across the three exam periods to assess how much they change over time and if time-varying covariate models may be more appropriate.

## Results

KM: * Sex: KM curves overlap throughout (no difference in survival, supported by log-rank Xˆ2 = 0.4, p = 0.5). * Diabetes: KM curves overlap mostly in the first 5 years (small numbers of early events). Diverge strongly later (log-rank: Xˆ2 = 34.3, p < 0.001). * BP: Strong visual difference on KM curves and log-rank (high BP has much higher stroke rate, separate cleanly). Xˆ2 = 149, p « 0.001.

Cox: * Males: diabetes and sysBP dominate the stroke risk, age and BP meds also contribute, CURSMOKE and TOTCHOL are marginal/borderline may not actually need to be included * Females: diabetes is a weaker predictor than for men, sysBP still dominates. Age, BP meds, and PREVCHD also contribute * PH tests look good overall. Only diabetes mildly violates for men (p = 0.008, more curvy), globally is good for each.

## Discussion

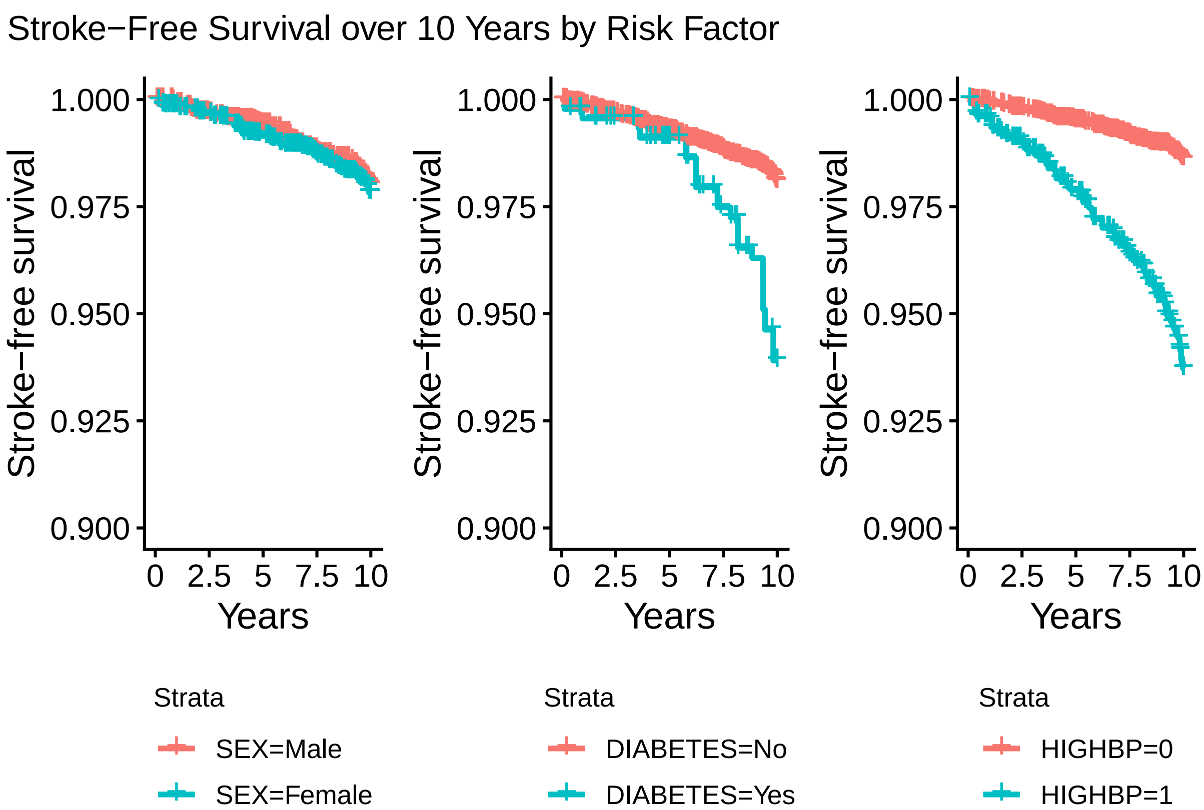
# Appendix

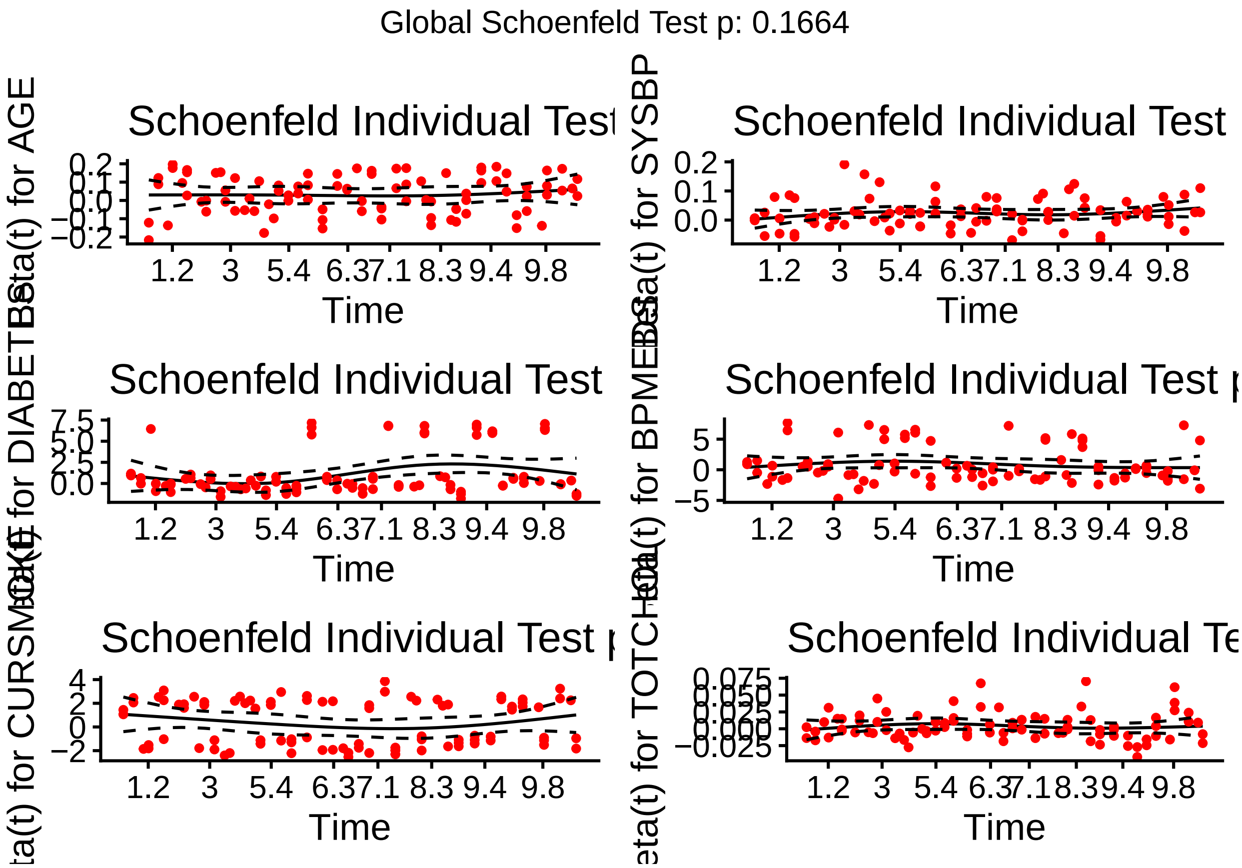
Code Directory: https://github.com/eweible/BIOS6624.git

File name: P3/Code/P3_Report.Rmd

Data Directory: C:/Users/weiblee/OneDrive/Documents/Anschutz/Spring 2026/6624_Advanced/Weible_Advanced/P3/DataRaw

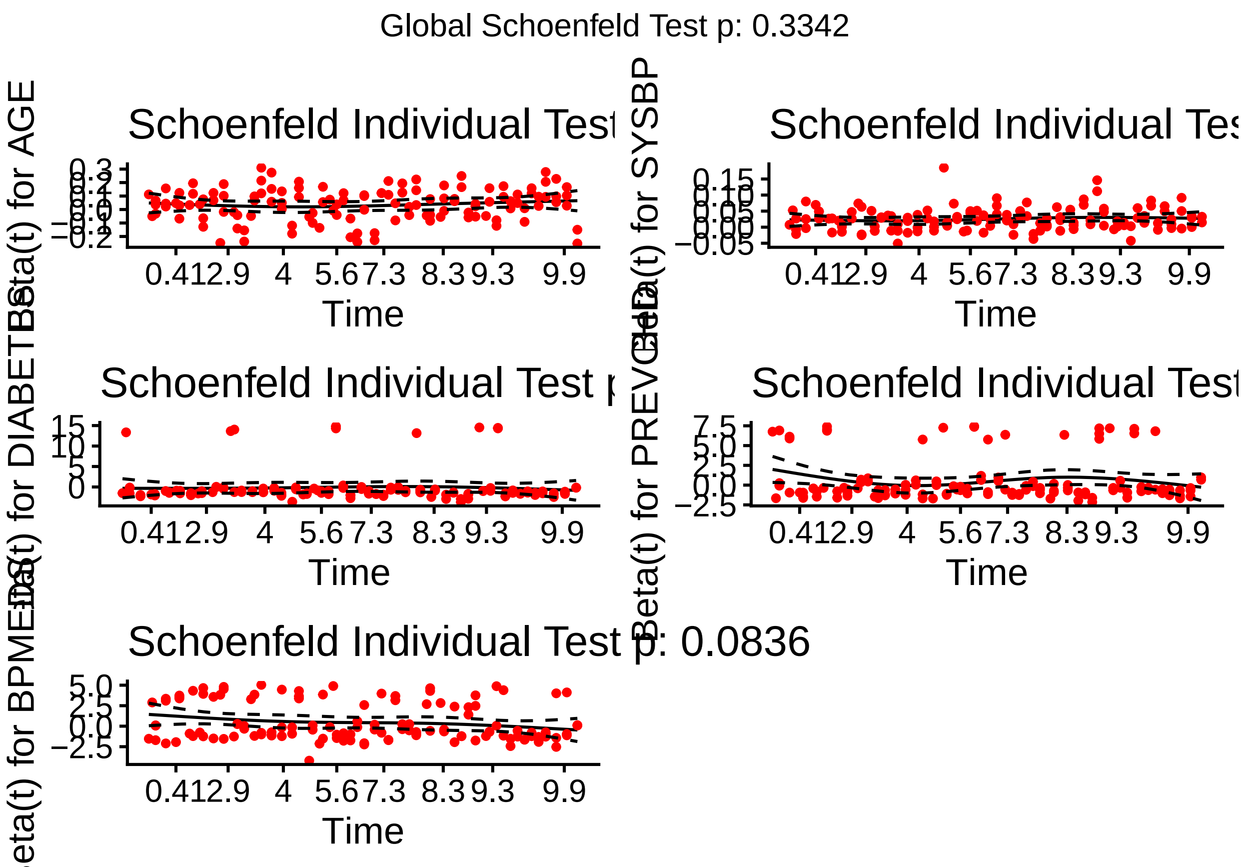
| | Male (N=4631) | Female (N=5911) | Overall (N=10542) |
|---|---|---|---|
| **AGE** | | | |
| Mean (SD) | 54.1 (9.45) | 54.6 (9.53) | 54.3 (9.50) |
| Median [Min, Max] | 54.0 [33.0, 80.0] | 54.0 [32.0, 81.0] | 54.0 [32.0, 81.0] |
| **SYSBP** | | | |
| Mean (SD) | 135 (20.1) | 137 (24.3) | 136 (22.6) |
| Median [Min, Max] | 131 [83.5, 232] | 132 [83.5, 295] | 132 [83.5, 295] |
| **DIABETES** | | | |
| No | 4405 (95.1%) | 5684 (96.2%) | 10089 (95.7%) |
| Yes | 226 (4.9%) | 227 (3.8%) | 453 (4.3%) |
| **PREVCHD** | | | |
| No | 4191 (90.5%) | 5629 (95.2%) | 9820 (93.2%) |
| Yes | 440 (9.5%) | 282 (4.8%) | 722 (6.8%) |
| **BPMEDS** | | | |
| No | 4369 (94.3%) | 5304 (89.7%) | 9673 (91.8%) |
| Yes | 262 (5.7%) | 607 (10.3%) | 869 (8.2%) |
| **CURSMOKE** | | | |
| No | 2203 (47.6%) | 3727 (63.1%) | 5930 (56.3%) |
| Yes | 2428 (52.4%) | 2184 (36.9%) | 4612 (43.7%) |
| **TOTCHOL** | | | |
| Mean (SD) | 235 (42.2) | 247 (47.1) | 242 (45.4) |
| Median [Min, Max] | 232 [113, 696] | 243 [112, 638] | 239 [112, 696] |

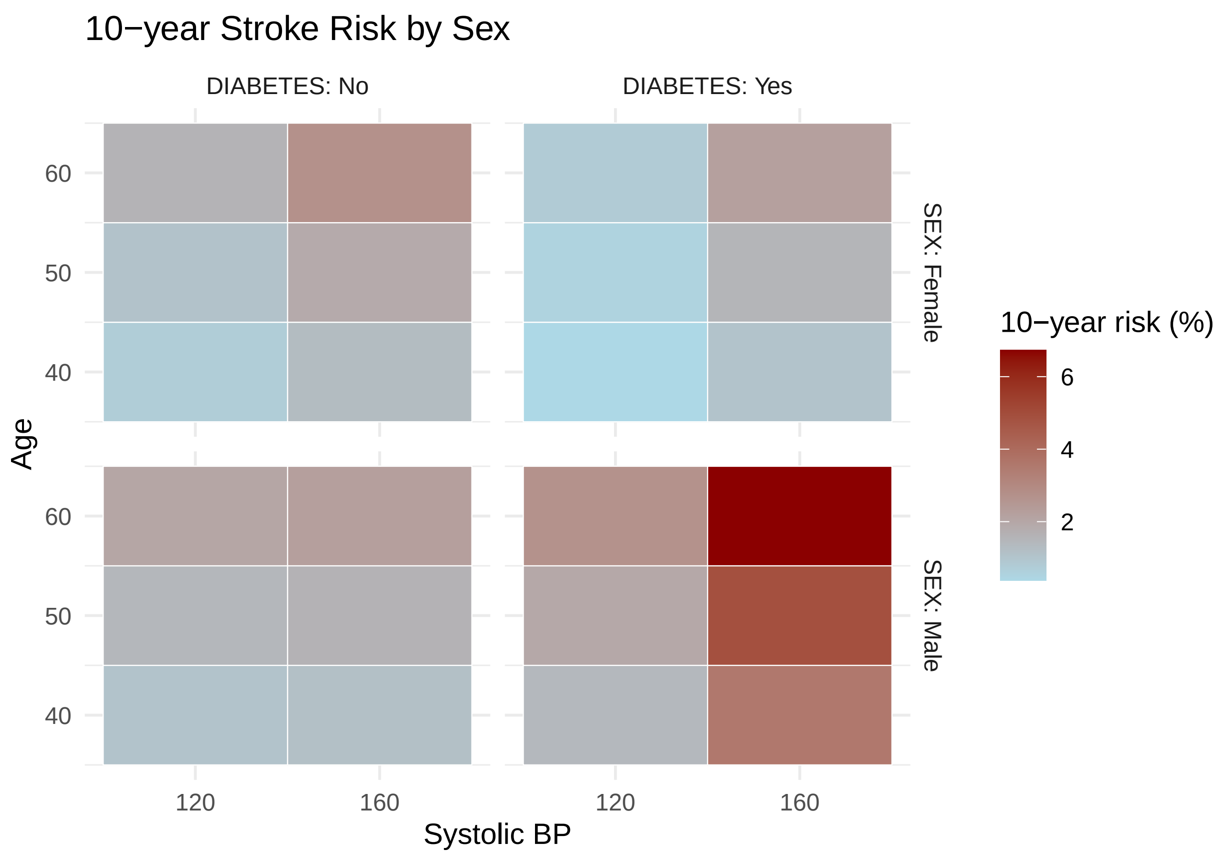

Stroke−Free Survival over 10 Years by Risk Factor
Stroke−free survival
1.000
0.975
0.950
0.925
0.900
0
2.5
5
7.5
10
Years
Strata
SEX=Male
SEX=Female
DIABETES=No
DIABETES=Yes
HIGHBP=0
HIGHBP=1

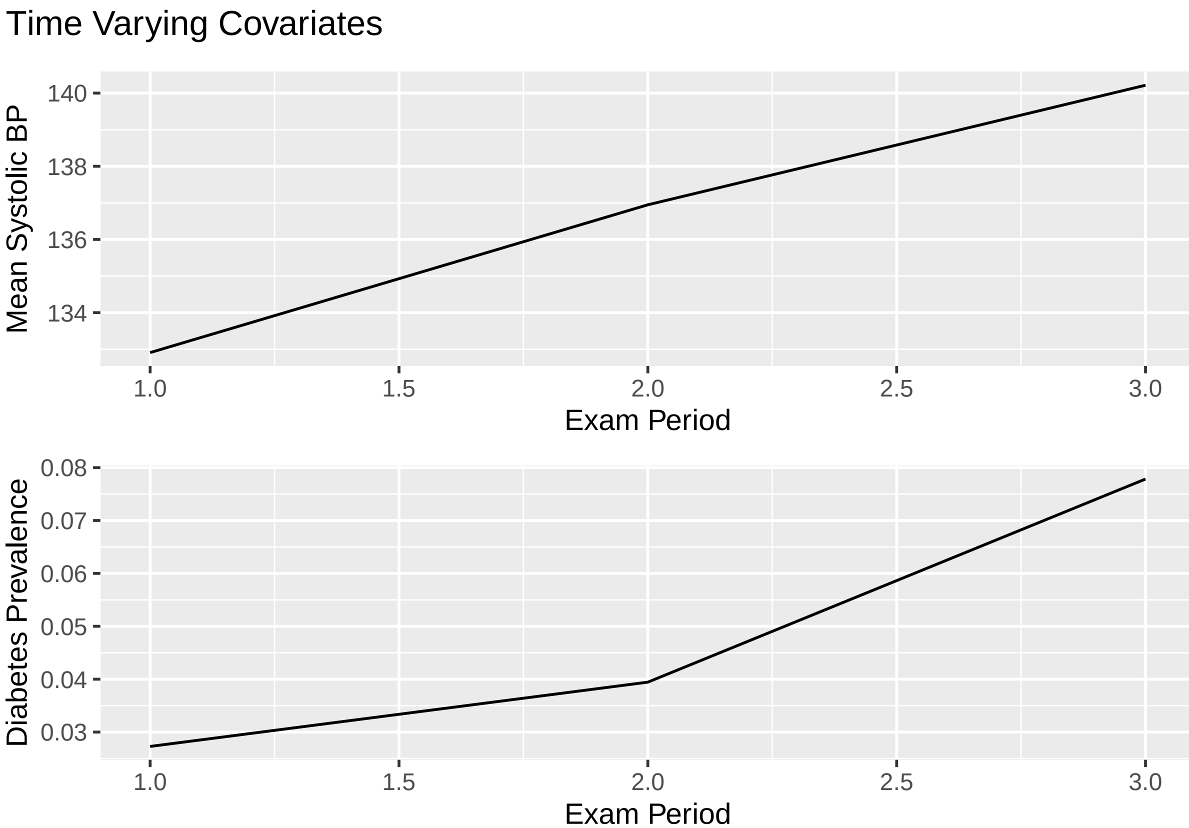

Global Schoenfeld Test p: 0.1664
Schoenfeld Individual Test
Beta(t) for AGE
Schoenfeld Individual Test
Beta(t) for SYSBP
Schoenfeld Individual Test
Beta(t) for DIABETES
Schoenfeld Individual Test
Beta(t) for BPMEDS
Schoenfeld Individual Test
Beta(t) for CURSMOKE
Schoenfeld Individual Test
Beta(t) for TOTCHOL
Time

# 10-year Stroke Risk by Sex and Clinical Profile

| | No Baseline Risk Factors | High Blood Pressure | Diabetes | High BP and Diabetes |
|---|---|---|---|---|
| Female | | | | |
| 40.0 | 0.5 | 1.3 | 0.4 | 1.1 |
| 50.0 | 0.7 | 1.9 | 0.6 | 1.5 |
| 60.0 | 1.0 | 2.7 | 0.8 | 2.2 |
| Male | | | | |
| 40.0 | 0.5 | 1.2 | 1.4 | 3.6 |
| 50.0 | 0.7 | 1.6 | 2.0 | 4.9 |
| 60.0 | 0.9 | 2.3 | 2.7 | 6.7 |




10−year Stroke Risk by Sex
DIABETES: No
DIABETES: Yes
SEX: Female
SEX: Male
Age
60
50
40
120
160
Systolic BP
10−year risk (%)
6
4
2


Time Varying Covariates
Mean Systolic BP
140
138
136
134
1.0
1.5
2.0
2.5
3.0
Exam Period
Diabetes Prevalence
0.08
0.07
0.06
0.05
0.04
0.03
1.0
1.5
2.0
2.5
3.0
Exam Period